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More than the Quantity: The Value of Editorial Reviews for a User-Generated Content Platform

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Abstract. We investigate an editorial review program for which a review platform supplements user reviews with editorial ones written by professional writers. Specifically, we examine whether and how editorial reviews influence subsequent user reviews (reviews written by noneditor reviewers). A quasiexperiment conducted on a leading review platform in Asia, based on several econometric and natural language processing techniques, yields empirical evidence of an overall positive effect of editorial reviews on subsequent user reviews from the platform's perspective. First, more reviews are provided for restaurants that receive editorial reviews. In addition, these reviews discuss substantive topics while also including a discussion on other topics, leading to a net increase in content length and variety. They also are more neutral in sentiment and are associated with lower rating valences. Further analysis of the mechanism reveals that the subsequent user reviews of the restaurants that receive editorial reviews become more similar to the editorial reviews in regard to topics, sentiment/rating, length, and readability, indicating a herding effect in how to write a review as the main driver of the change in the subsequent reviews. We further empirically isolate this herding effect among long-time reviewers. The findings suggest that review platforms could use an editorial review program not only to boost the quantitative aspect of user reviews but also, to manage the qualitative aspect as well.

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1. Introduction

Online user-generated content (UGC), such as consumer reviews, plays an important role in the decision-making process of shoppers and thus, creates a direct economic impact (e.g., Dellarocas 2003). With the advent of technology, review platforms have benefited from a “wisdom of the crowd” effect, whereby peer consumers provide a significant amount of feedback. Not surprisingly, platform owners have employed various strategies to improve the quantity and quality of the review content, which has included offering extrinsic incentives (Khern-am-nuai et al. 2018), using intrinsic social norms (Burtch et al. 2017), or creating social networks (Goes et al. 2014). In addition, one of the techniques that has gained traction in recent years among online review platforms is to incorporate expert reviews. These reviews are generated by the third-party experts who are neither directly associated with the seller nor existing consumers, such as platform-hired

professional editors/testers (e.g., Qunar.com), solicited guest testers from platform users who are recognized by their past expert-quality review generation (e.g., Amazon, Yelp, Dianping, FlyerTalk, Taobao, JD.com), or contracted experts/critics in the domain (e.g., Rotten Tomatoes).

In this study, we focus on one specific type of such expert review programs, namely editorial review. As compared with other expert review programs (e.g., the invited critics on Rotten Tomatoes) that usually allow expert reviewers/critics to have a certain degree of freedom (i.e., how to review and which product to review), the editorial review program tends to be more restrictive, whereby expert reviewers serve as the editors affiliated with and managed by the platforms. It endows the platform with additional controls regarding how to organize the review content and to which product to provide such reviews. In practice, platforms that employ the editorial review program

typically hire and train editors internally (e.g., Qunar's reviews by professional testers) or selectively invite high-quality reviewers as guest editors and solicit them to review targeted merchants/products/services through invitation-based and complimentary/paid testing programs (e.g., Amazon Vine Program; elite programs of Yelp and FlyerTalk; invited testing programs of Dianping, Taobao, and JD.com). Reviews generated under such an affiliation/management relationship are usually displayed with an explicit label, such as "editorial review" or "invited review," on the review page.

Theoretically, there exist competing tensions that may influence the implications of editorial reviews on the platform. On the one hand, these editorial reviews may augment the content of the review platform by providing discussion points and thus, stimulate user participation. On the other hand, editorial reviews also may unintentionally discourage users by triggering substitution effects (i.e., deter users from contributing similar reviews) or by being perceived as marketer-generated content. In this research, we resolve these competing tensions by investigating empirically how the presence of editorial reviews affects subsequent consumer reviews both quantitatively and qualitatively.

Our research agenda is important and relevant to research and practice. First, although the impact of existing user-generated reviews on subsequent reviews has been well studied (e.g., Schlosser 2005, Moe and Trusov 2011, Moe and Schweidel 2012, Lee et al. 2015), these results may not be directly applicable to analyzing the implications of editorial reviews, as consumer reviews and editorial reviews are vastly different. For example, although editorial reviews, in general, are of high quality because experts have incentives to keep their reputation, this is usually not the case for consumer reviews (Burtch et al. 2017). In addition, existing research has demonstrated that review readers may react to these two types of reviews differently (Goh et al. 2013). As a result, editorial reviews are expected to have a different impact than consumer reviews on subsequent reviews, which is an issue that has not been studied in the literature.

To operationalize our research agenda, we collaborate with a large restaurant review platform in Asia to study the impact of editorial reviews. We leverage a quasiexperiment research design in which our focal review platform launched the editorial review program while everything else remained the same. We first examine the impact of editorial reviews on basic characteristics of subsequent user-generated reviews, including volume, length, and valence. Next, we incorporate natural language processing (NLP) methods to examine textual content of subsequent user reviews and to quantify the impact of editorial reviews on the

content-related features of those reviews. We then leverage the theory of herding and differentiation behavior to explain and examine the mechanism of editorial reviews that drives the change in users' review-generating behavior. In a further examination, we develop similarity measures to compare the review-level content differences from multiple dimensions between editorial reviews and user reviews before and after the existence of editorial reviews. Finally, we develop a feasible identification strategy that leverages long-term users of the platform to isolate the change in their review-writing behavior that is driven by editorial reviews.

Our analyses reveal several notable findings. First, restaurants that receive editorial reviews enjoy an increase in review volume afterward, suggesting herding behavior from platform users toward editorial reviews in choosing which restaurants to review. In addition, in regard to the review content, user reviews posted after editorial reviews are longer and have a higher degree of content variety because of increased substantive content, such as discussions on food, desserts, and drinks. Notably, the topic distribution of user reviews gradually gravitates toward that of editorial reviews. Further, the existence of editorial reviews greatly affects the sentiment of subsequent user reviews as well. Specifically, user reviews become more neutral and are associated with lower rating valences after the editorial reviews are posted, also exhibiting gravitational effects toward the sentiment of editorial reviews. In this regard, our review-level mechanism analysis of similarity measures demonstrates quantitatively that the content of user reviews becomes similar to that of editorial reviews in all aspects (i.e., topic, sentiment/rating, length, and readability). In other words, with the editorial reviews present, review writing on the platform gradually tends toward the content of editorial reviews. We also conduct a reviewer-level analysis that leverages the long-term reviewers. Such an alternative identification strategy confirms that editorial reviews trigger the behavioral changes of the reviewers, suggesting the existence of a herding effect (i.e., users follow professional editors) in their review-writing behavior.

Our study offers several key contributions to the literature and practice. First, it contributes to the literature stream on herding or differentiation effects in the context of UGC. Although prior research has thoroughly examined herding and differentiation among users, we further explore the effects between users and platform editors, which helps to improve our understanding of review-generating behaviors. Given that it is particularly difficult for review platforms to manage the natural behavior of users, our work provides tangible practical insights on the herding behavior that occurs around the content generated by the platform. Second, this research contributes to the

literature on third-party expert reviews by investigating how they affect word of mouth (WOM) in the format of consumer reviews. Because previous literature has demonstrated empirically a significant economic impact of WOM (e.g., Chevalier and Mayzlin 2006), insights on the impact of editorial reviews on WOM in multiple dimensions could be extrapolated to identify a potential indirect influence on the economic outcomes, which is increasingly important with the prevalence of digitized UGC and electronic word of mouth (eWOM). Third, our results yield important insights for managers of UGC platforms. Although numerous studies have examined the effectiveness of various managerial strategies to improve UGC, such strategies usually target either content quantity (e.g., Zhang and Zhu 2011) or content quality (e.g., Ransbotham et al. 2019). Some of these strategies may increase the quantity at the expense of quality (e.g., Khern-am-nuai et al. 2018, Burtch et al. 2022). Our study considers editorial reviews as a viable strategy to manage UGC and shows their effectiveness not only in the quantitative aspects but also, in the qualitative aspects as well. The insights of editorial reviews could help platform managers to derive content management policies that contribute to the platform's long-term viability and sustainability.

The remainder of the paper is organized as follows. In Section 2, we survey the literature related to our study. In Section 3, we describe the empirical context of this study, along with our data set. In Section 4, we present the identification strategy and empirical models. In Section 5, we provide the results and findings. We conduct additional robustness checks, which are included in Section 6. Finally, in Section 7, we discuss the theoretical and practical contributions of this study, the limitations, and future research opportunities as well as conclude the paper.

2. Literature Review

In this section, we survey the literature that is closely related to this study. In particular, we review previous works that examine herding and differentiation behavior. We then discuss studies that investigate third-party evaluation, followed by those that focus on how review platforms manage UGC.

2.1. Herding and Differentiation Behavior

Herding effects, known as an individual's tendency to follow actions taken by previous decision makers, have been studied empirically in several contexts, including microloan markets (Zhang and Liu 2012), restaurant dining (Cai et al. 2009), and online flash sales (Cui et al. 2019). In those contexts, herd behavior is shown to be a consumer choice (i.e., whether to choose a product or which product to choose), which is then aggregated to changes in sales volume.

Herding behavior is also known as the bandwagon effect (Marsh 1985), whereby individuals tend to choose an option chosen more often by prior individuals. In the specific context of UGC, herding behavior is used to explain how existing UGC affects subsequent content generation. For example, Schlosser (2005) finds that reviewers tend to decrease their product evaluations if they observe negative opinions expressed by others. Similarly, Moe and Schweidel (2012) show a bandwagon effect in reviewers' writing behavior; reviewers tend to conform to the majority by adjusting their product evaluations upward in more positive review environments. Such herding effects are found to be especially prominent when reviewers share close social ties (Lee et al. 2015).

Conversely, differentiation effects are the tendencies of individuals to distinguish themselves from others. In the context of UGC, differentiation effects are used to explain the cases in which reviewers adjust their product evaluations such that they are different from those of others. For instance, Moe and Trusov (2011) provide empirical evidence that some reviewers strive to differentiate their reviews from existing ones. Likewise, Schlosser (2005) argues that self-presentation concerns lead reviewers to differentiate their reviews. Previous works have attributed such behavior to reviewers' "need for uniqueness" (Cheema and Kaikati 2010, Zeng and Wei 2013). These differentiation effects are strongly related to disconfirmation behavior where consumers use prior evaluations to set their expectations (e.g., McKinney et al. 2002, Ho et al. 2017). In addition, differentiation effects influence consumer decisions on whether to contribute reviews by discouraging them from reviewing products with high review volume (Moe and Schweidel 2012). Such an effect occurs because new receivers perceive that their opinions could be inconsequential because of the number of existing reviews.

Although past research has thoroughly examined herding and differentiation in the context of UGC among end users, it is particularly difficult to derive practical insight into how to leverage these effects to improve review platforms, as review platforms cannot directly manage consumer reviews or intervene in consumers' review-writing processes to take advantage of natural herding/differentiation behavior. In this regard, we extend this stream of literature by exploring users' herding or differentiation behavior with respect to platform editors, who are more subject to the platform's control. In particular, we examine whether the content of editorial reviews could influence the content-generating behavior of users, and we use the herding or differentiation effect to explain the underlying mechanisms. By doing so, we provide review platforms with practical implications

for utilizing social influence theory, as platforms could centrally manage editorial reviews and use editorial reviews to stipulate users' content-generating behavior. Next, as editorial reviews are considered one of the third-party expert review programs, we review the literature on third-party evaluations.

2.2. Third-Party Evaluations

Traditionally, third-party evaluations often refer to expert evaluations conducted by a third party that have no official association with the party being accessed (Ozpolat et al. 2013). Most studies of third-party evaluations in this form, such as certification and licensing, examine their impacts on traditional market outcomes. For example, Jin and Leslie (2003) show that third-party evaluations can increase consumer demand and improve seller quality. Similarly, Tirole (1988) finds that third-party evaluations can help to reduce information asymmetries and thus, increase demand from buyers who otherwise may not have entered the market.

Our work connects to an emerging body of research that examines the role of third-party evaluations in an online review context. The literature has documented significant impacts of third-party reviews on consumer behavior. For instance, Eliashberg and Shugan (1997), Basuroy et al. (2003), and Chen et al. (2012) show that film critics can greatly affect the popularity of movies and can thus serve as a good predictor of movie box office revenue. Likewise, Reddy et al. (1998) find that newspaper critics have a significant impact on consumers' attitudes toward Broadway shows, with critics from popular newspapers such as *The New York Times* yielding much more influence than those featured in relatively unpopular newspapers. Lee and Tan (2013) show that third-party evaluations can effectively reduce users' difficulties in evaluating, which can lead to a higher product adoption rate. Further, Shaffer and Zettelmeyer (2002) show that, when the sentiment of third-party reviews affects consumers' product evaluations, the relevance between review topics and customer segments also plays a pivotal role in affecting consumer purchasing behavior. In this regard, Chen and Xie (2005) emphasize the importance of third-party reviews by showing that manufacturing firms should adapt their marketing strategies in response to third-party reviews.

Although prior studies have advanced our understanding of the effects of third-party evaluations on market outcomes (e.g., consumer demand), the implications and underlying mechanisms of third-party evaluations in the online review context remain understudied. Potentially, third-party reviews could have a salient influence on consumer-generated reviews. Indeed, the prevalence of online review platforms suggests an even stronger impact of third-party

reviews if the review exhibits an effect on eWOM, such as UGC. Such an effect warrants investigation. In this regard, our research examines empirically whether editorial reviews, a form of third-party reviews, influence subsequent consumer reviews in terms of content quantity and quality. As editorial reviews in the context of this research are under the control of a review platform, we survey the literature on the practice used by review platforms to manage UGC.

2.3. Review Platform's Management of UGC

Prior studies that have examined the review platform's management of UGC can be broadly classified into three substreams of research. The first substream concerns platform strategies to enhance the consumer's review-reading experience. For instance, many review platforms provide helpfulness information (i.e., helpful and unhelpful votes) alongside each review. This metric not only serves as a primary index for consumers to filter out low-quality reviews but also, sorts and positions the most helpful reviews, both positive and negative, more prominently on the review pages (Yin et al. 2016). In addition, because reviews written by opinion leaders are perceived as more influential than are others, many review platforms reward prolific reviewers with badges to signal their status to other users (Yazdani et al. 2018). Further, to reduce fake reviews, some review platforms (e.g., Expedia.com) allow only those customers who transact on the platform to post user reviews (Mayzlin et al. 2014), whereas other platforms (e.g., TripAdvisor) implement review filters to detect and eliminate suspicious content (Lappas et al. 2016).

The second research substream focuses on feasible avenues for review platforms to increase UGC production. Given the consistent findings of the positive effects of review volume on economic outcomes (e.g., Chevalier and Mayzlin 2006, Zhu and Zhang 2010, Moe and Trusov 2011), it is important for review platforms to understand the antecedents of UGC production and to develop mechanisms that foster reviewer participation. Studies show that the motivational factors for UGC production vary widely based on reviewers' characteristics. The examples of these factors are audience group size (Zhang and Zhu 2011), perceived identity verification (Ma and Agarwal 2007), and community commitment (Bateman et al. 2011). Based on these factors, several platform interventions are designed to stimulate UGC production. For instance, Cabral and Li (2015) suggest the use of financial incentives to increase the likelihood of feedback. Chen et al. (2010) show that the use of social norms (i.e., providing users with peers' contribution activity) can have a positive influence on the production of movie ratings. Online performance feedback interventions

also are shown to potentially drive UGC contributions (Huang et al. 2019).

Third, prior works have investigated possible methods that review platforms can employ to improve the quality of review content. In particular, researchers have investigated the antecedents of UGC’s qualitative characteristics on several dimensions, such as content length and linguistic features (e.g., sentiment, readability). For example, Goes et al. (2014) find that allowing a subscription or “following” could be an effective strategy to improve review content because the contributor’s popularity (e.g., number of subscribers) is shown to be correlated with the readability of the reviews. Zeng and Wei (2013) support this strategy by showing that social ties (e.g., social network neighbors) can increase the similarities between the content that people create online. Further, review platforms could use the combination of financial incentives and social norms (Burtch et al. 2017) or performance-contingent monetary incentives (Yu et al. 2022) to motivate reviews of greater length.

Our research complements this body of literature by exploring the strategic value of editorial reviews from the platform’s perspective. To examine the effectiveness of this strategy, we investigate its effects not only in the quantitative aspects but also, in the qualitative aspects as well. In particular, we study the impact of editorial reviews on subsequent user review volume, rating, length, topics, sentiment, and readability. Further, we reveal the underlying mechanisms

for the change of reviewers’ writing behaviors that are directly influenced by editorial reviews.

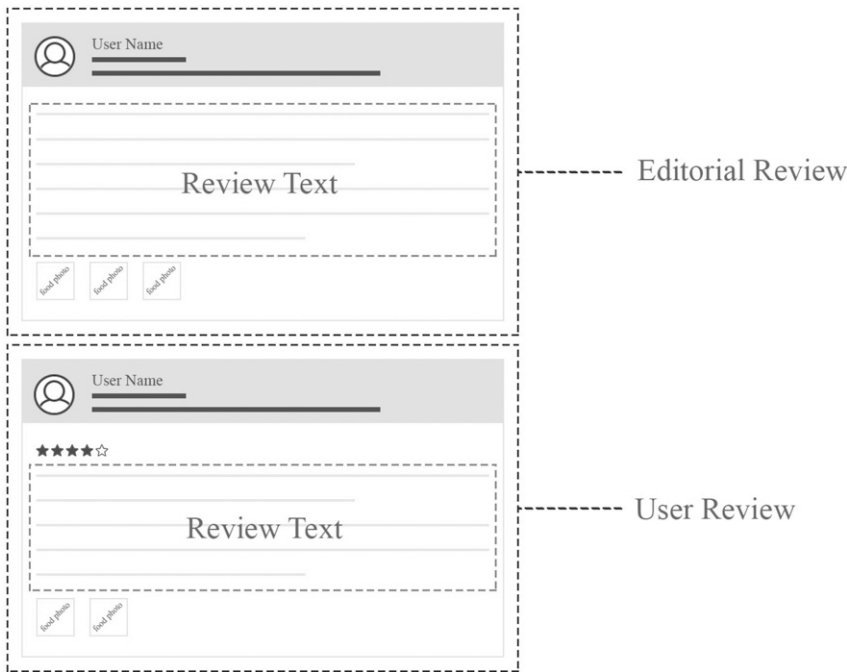
3. Research Context and Data

In this section, we present the empirical context of our study and discuss the descriptions of our data and variables of interest.

3.1. Research Context

In this research, we collaborate with a large restaurant review platform in Asia that provides services that are similar to those of other major review platforms in the United States, such as Yelp. The design of its website with anonymized details is provided in Figure 1. Each restaurant on the platform has a dedicated page that contains detailed information about the restaurant (e.g., address, menus, opening hours) as well as reviews and ratings contributed by platform users, sorted by posting date (i.e., the most recent review is displayed on the top of the list). Users who register on the platform with a valid email address can submit reviews and ratings. A typical review generated by users consists of a restaurant rating (on a scale of one to five), the textual content of the reviews, and any relevant photos. The platform also features a peer evaluation system that allows users to vote for other reviews. All reviews are stored and published by the review platform.

Figure 1. User Review and Editorial Review Examples



In addition to user-generated reviews, the platform started to provide reviews written by professional editors (i.e., editorial reviews) in late 2010. The editorial review program was launched to serve various purposes. First, the platform recognizes that user-generated reviews alone tend to suffer from reporting bias (Hu et al. 2007, Pamuru et al. 2021). Therefore, consumers who read user-generated reviews may experience some disparities between the actual restaurant quality and the perceived quality based on the reviews. Hence, the platform intends for the editorial reviews to be objective and comprehensive evaluations of restaurant quality to balance the sentiment in user-generated reviews. Second, the comprehensiveness of editorial reviews could account for features, topics, or aspects of the restaurants that are missing or rarely discussed in user-generated reviews. In the same way, the editorial reviews could reduce noise that occurs in the user-generated reviews.

To ensure the quality of editorial reviews, the editors hired by the platform are professional editors/critics who have received sufficient training on the expectations for their contributions. In addition, before posting editorial reviews, the platform crosschecks each editorial review to ensure that the content satisfies the platform's editorial review standard (i.e., that the review is reasonably objective and comprehensive). As we will demonstrate in the next subsection, the summary statistics of posted editorial reviews confirm the effectiveness of such practices. Specifically, editorial reviews are comprehensive in the covered topics, as they tend to substantially discuss food-related aspects (e.g., food courses, desserts, drinks). In addition, they are neutral in sentiment and of a substantial length (see Table 4 for details).

Because of the limited number of professional editors, the platform is unable to provide editorial reviews for all restaurants. The platform managers are tasked to select restaurants to receive editorial reviews based on their business characteristics, including popularity, food category, price range, region, and so forth. Notably, the platform also favors restaurants that sign advertising contracts with the platform to receive editorial reviews.¹ Subsequently, the managers assign available professional editors to visit those restaurants as ordinary consumers and to write corresponding editorial reviews after the visit. In total, 2,283 of 145,141 restaurants on the platform have received editorial reviews.

The format of editorial reviews is generally consistent with that of user-generated reviews, as shown in Figure 1. Specifically, both types of reviews offer some generic information in regard to reviewers (e.g., screen name, profile picture), and the detailed descriptions of the restaurants are provided as textual content. Nevertheless, there are some differences between

the two. First, editorial reviews are clearly marked as “editorial.” Second, no numeric ratings are associated with the editorial reviews, as the platform aims to provide reviews with objective evaluations. Third, editorial reviews, if they exist, are placed on the top of the review pages (i.e., above all user-generated reviews). Note that one restaurant may receive more than one editorial review, which may or may not be written by the same editor. In that case, only the most recent one appears at the top, whereas the former ones will be pooled with user reviews and sorted chronologically as of the posting time.

3.2. Data

In this subsection, we present the data set used in this study. We begin by describing the data and variables of interest. Then, we explain NLP techniques that we apply to the textual content in our data set.

3.2.1. Data Description and Variable Definitions.

Through the collaboration, we obtained access to the complete review-level data of 145,141 restaurants, including all user-generated reviews from July 1, 2010 to May 30, 2017 and all editorial reviews during this period. Restaurants in the data set started to receive editorial reviews in late 2010. We check that none of the restaurants have an editorial review as its first review (i.e., before any user reviews). The data set contains the information that pertains to online reviews, including the review's textual content, date, rating, and so forth. Further, we have access to the generic information for all restaurants on the platform, including whether a restaurant has signed an advertisement contract with the platform. Using this data set, we conduct two sets of analyses in our study.

First, we focus on simple review characteristics that are not embedded in the textual content. Following Chevalier and Mayzlin (2006), we examine three basic review characteristics: (1) *ReviewVolume*, (2) *Rating*, and (3) *Length*. We construct our data set as panel data at the restaurant level such that each observation is a restaurant and each time period is a month. Therefore, $ReviewVolume_{it}$ denotes the total number of user reviews that are posted for the restaurant i in month t . $Rating_{it}$ represents the average rating of reviews for restaurant i in month t . $Length_{it}$ is the average number of review characters for restaurant i in month t .

Second, to better capture characteristics of the content of reviews, we extend our study beyond the surface of the aggregated review characteristics by focusing on the textual content. Here, the unit of analysis becomes the review level. We follow Ghose and Ipeirotis (2011) to extract three major qualitative metrics from the textual content of each review (editorial and user-generated reviews) by using NLP techniques. The qualitative metrics extracted are (1) topics covered in reviews, (2) review

sentiment (i.e., the fraction of negative, positive, or neutral emotion in the review text), and (3) readability (e.g., Simple Measure of Gobbledygook (SMOG) index). We provide more details on the construction of these variables in Section 3.2.2.

The independent variables and control variables are tailored to the two aforementioned levels of analyses. Our main independent variable of interest is the treatment indicator, $EditorialReview_{i(k)t}$, which indicates whether an editorial review has been provided for restaurant i by (the time when posting the k th user review in) month t . Similarly, our control variables include

- $Ad_{i(k)t}$: whether restaurant i is in an advertisement contract with the review platform by (the time when posting the k th user review in) month t ;
- $PriorVol_{i(k)t}$: the cumulative user review volume for restaurant i by (the time of posting the k th user review in) month t ;
- $PriorERVol_{i(k)t}$: the cumulative editorial review volume for restaurant i by (the time of posting the k th user review in) month t ; and
- $PriorRate_{i(k)t}$: the cumulative rating valence for restaurant i by (the time of posting the k th user review in) month t .

We also include the reviewer experience of the k th user review in month t , $Experience_{ikt}$, measured as the duration between t and his or her first posting time (in month) as a control variable in the mechanism analysis. We further collect various static restaurant characteristics, including product tags, number of available payment methods, access to free parking, number of seats static, food category, city, and price range, to augment our identification strategy, which we explain in Section 4.

3.2.2. Natural Language Processing for Review Text.

Because the review text may contain additional information that is missing from aggregated variables, such as review volume and valence (Archak et al. 2011), we apply the following NLP techniques to better extract the qualitative characteristics from the text content of editorial reviews and user reviews. Recall that all analyses in this subsection are performed at the review level.

3.2.2.1. Topics. First, we utilize topic modeling to discover hidden semantic structures in the textual content of the reviews in our data set. For each review, we find the abstract topics through the following steps. First, we apply the data preprocessing techniques of tokenizing to delete meaningless stop words (e.g., “a,” “and,” “the”) and stemming to convert our raw review data into a clean data set. Second, making use of term frequency-inverse document frequency, we vectorize each review based on the extracted

keywords in the context of restaurant reviews. Third, we use latent Dirichlet allocation (LDA) (Blei et al. 2003) as our topic model to identify the latent topics mentioned in our review data and to obtain the proportion of each latent topic in each review. Note that, in the topic modeling process, we pool editorial reviews and user reviews together for the following reasons. First, by pooling both types of reviews together, LDA results provide us with a unified Euclidean space that makes any editorial reviews and user reviews comparable. This structure is particularly important for our mechanism analysis, for which we measure the similarity between an editorial review and a user review. Second, pooling provides additional variations (i.e., the variations between editorial reviews and user reviews) for LDA to uncover more topics with which user reviews are distinct from editorial reviews in addition to the topics that differentiate one review from the others within user reviews or editorial reviews.

To determine the number of topics, we develop a performance metric based on the coherence of the words in each topic and the separation between each topic, following Griffiths and Steyvers (2004), Cao et al. (2009), and Zhao et al. (2015). A smaller value of the metric suggests better performance of the model. We then vary the number of LDA topics from 2 to 10 and calculate the metrics for each value. The performance metrics identify five topics as the optimal choice, as reported in Table 1. Using this model, we list the major contributing words in each topic and manually label the latent topics, as shown in Table 2. The five topics are (1) *Food Course*, (2) *Dessert & Drink*, (3) *Emoji*, (4) *Service*, and (5) *Atmosphere*. In addition, topic modeling allows us to represent the topics of each review by using a topic vector $T_{ikt} = [t_1, t_2, \dots, t_5]_{ikt}$ where $\sum_{j=1}^5 t_j = 1$. Specifically, t_j (where $j = 1, 2, \dots, 5$) in this vector represents the percentage of topic j in the k th review of restaurant i in month t . For example, if a topic vector of a certain review is $\{0.7, 0, 0.2, 0.1, 0\}$, this indicates that the topic in regard to *Food Course* dominates in this review, whereas this review also includes some *Emoji* and a few descriptions of *Service*.

3.2.2.2. Sentiment. Sentiment analysis is an NLP technique that extracts review sentiment in a quantitative manner. The primary objective of our sentiment analysis is to determine the perception of a reviewer with respect to the restaurant while writing reviews.

Table 1. Determination of the Number of Topics

Topics	2	3	4	5	6	7	8	9	10
Performance metrics	0.445	0.434	0.382	0.346	0.374	0.415	0.443	0.548	0.571

Table 2. Most Contributing Words for the Five Latent Topics in Reviews

Topic	Most contributing words
1 (<i>Food Course</i>)	Pork, shrimp, fish, chicken, rice ...
2 (<i>Dessert & Drink</i>)	Cream, sweet, ice, tea, milk ...
3 (<i>Emoji</i>)	U + 1F60x, U + 1F92x, U + 1F9Ex ...
4 (<i>Service</i>)	Wait, service, staff, time, people ...
5 (<i>Atmosphere</i>)	Atmosphere, nice, air, friend, environment ...

Specifically, we conduct the sentiment analysis to classify the polarity of each review text by quantifying whether the expressed opinion in the review text is positive, negative, or neutral (Pang and Lee 2008). In terms of sentiment analysis algorithms, we implement a rule-based approach that is based on the lexicon Valence Aware Dictionary and sEntiment Reasoner (VADER), which is a sentiment analysis package developed primarily for social media text (Hutto and Gilbert 2014). This analysis enables us to label each review with three sentiment scores (i.e., positive score S^{pos} , negative score S^{neg} , and neutral score S^{neu}), where the summation of three scores equals 1 (i.e., $S^{pos} + S^{neg} + S^{neu} = 1$), and each score is nonnegative. S^{neu} rates the neutral assertions that exist in a review. Similarly, S^{pos} and S^{neg} represent the proportions of positive and negative assertions, respectively. When a particular score is greater than the other two, this indicates that the tone of this review is driven toward a certain polarity. For instance, if the sentiment scores $\{S^{pos}, S^{neg}, S^{neu}\}$ of a certain review are $\{0.8, 0, 0.2\}$, respectively, we can infer that this review is positive.

3.2.2.3. Readability. The concept of readability analysis refers to the ease with which a reader can understand the written textual content. The readability score is calculated by the complexity of vocabulary and syntax. In the field of NLP, various readability specifications are available for an overall readability score evaluation. We use the SMOG formula to determine the years of education needed to understand the review text, following prior literature (e.g., Ghose and Ipeirotis 2011). By using the number of polysyllabic words, we calculate the SMOG score for each review. A higher SMOG score indicates that the review is more difficult to understand. For example, if the SMOG score of a certain review equals six, it indicates that six years of education are required to fully understand this review.

3.3. Model-Free Evidence

In this subsection, we visualize the changes in user reviews after the restaurants receive editorial reviews as model-free evidence, as shown in Figure 2.

We find that, as compared with user reviews before editorial reviews, those written after editorial reviews

have left-shifted ratings and positive sentiment scores as well as right-shifted lengths, neutral sentiment scores, and topic density on *Food Course* and *Dessert & Drink*, whereas there are unclear changes for other characteristics. This suggests that reviews after the editorial reviews tend to have lower ratings; less positive sentiments; more neutral sentiments; increased length; and more discussions on food courses, desserts, and drinks. Moreover, almost all review characteristics (except review volume) exhibit thinner distributions after editorial reviews, suggesting a homogenizing tendency.² These findings, however, reveal only the before and after differences in the user reviews and may be confounded by several factors. We next present our econometric analyses on the impact of editorial reviews.

4. Identification Strategy

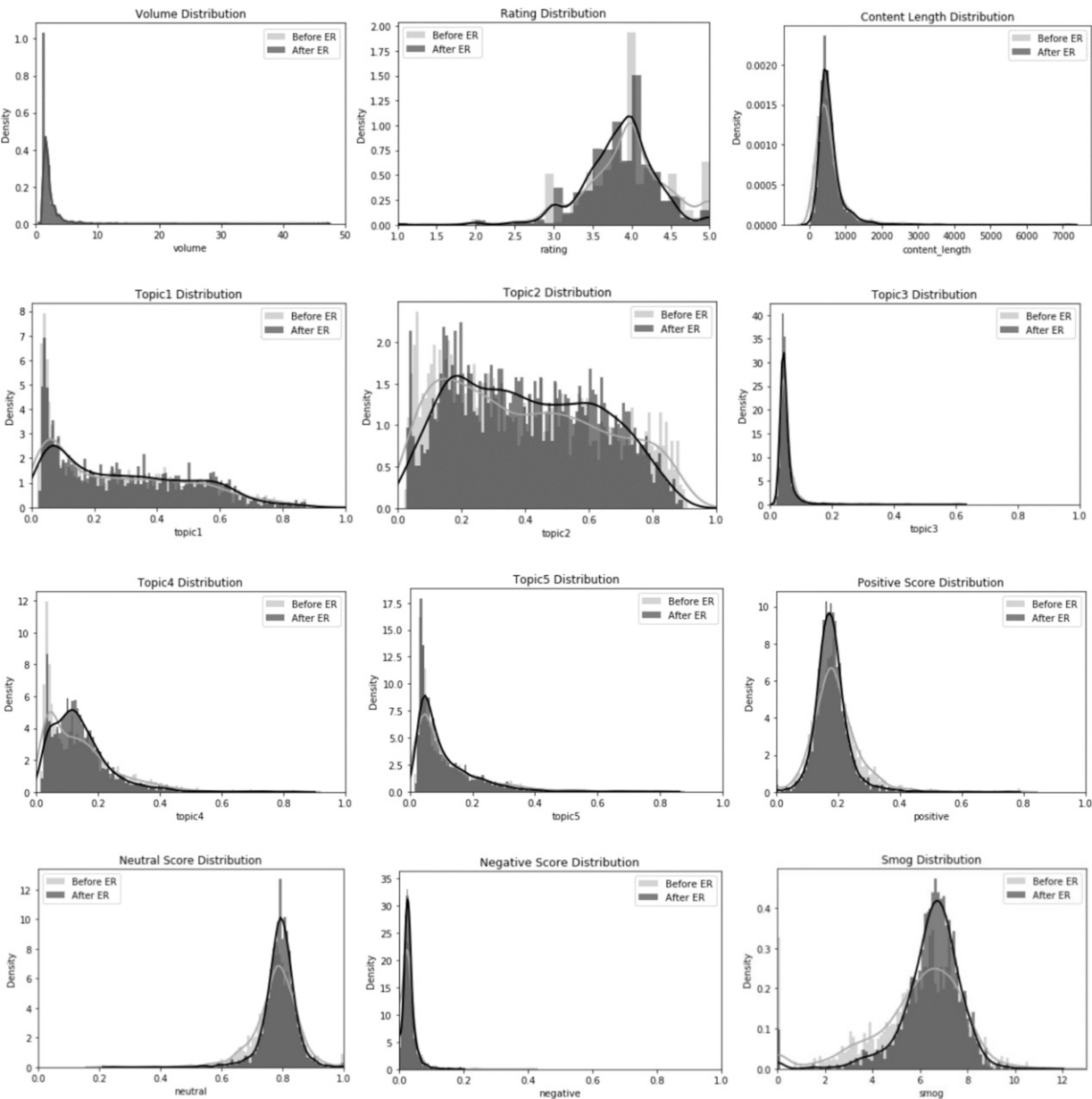
In this section, we describe our identification strategy, which depends primarily on the combination of propensity score matching (PSM) and difference-in-differences (DID) regression specification.

4.1. Main Specification

When identifying the potential effects of editorial reviews, it is important to note that restaurants that receive editorial reviews may be significantly different from restaurants that do not receive them. Therefore, we utilize the matching technique to create a group of restaurants that receive editorial reviews (“treatment group”) and a group of restaurants that are similar to the ones in the treatment group but do not receive editorial reviews (“control group”). Using matching, we structure our data to mimic a controlled experiment, which allows us to tease out the causal effects of editorial reviews. Following that, we use the DID technique as the main regression specification. This approach is typically used to estimate the effect of an exogenous treatment by comparing the changes in outcomes over time between the treatment and control groups (e.g., Chan and Ghose 2013, Khern-am-nuai et al. 2018). DID is suitable in this research context because we observe a fundamentally quasiexperiment design, whereby the treatments (i.e., editorial reviews) are applied to different businesses at different time points that have user reviews both before and after, whereas many other restaurants do not receive this treatment. In particular, for treatment restaurants, user reviews will be affected by editorial reviews if they are posted after editorial reviews. For control restaurants, there are no such effects.

4.1.1. Matching. As previously noted, the platform selects restaurants to receive editorial reviews based on multiple business characteristics. Thus, restaurants that receive editorial reviews may be different from those that do not receive editorial reviews. Such a

Figure 2. Review Characteristics Before and After Editorial Reviews (ERs)



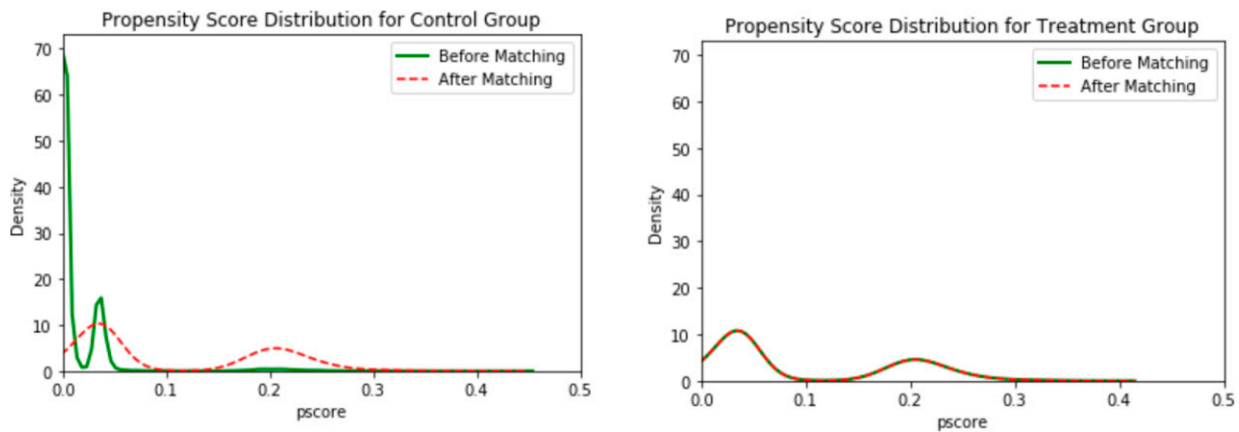
selection bias could confound our results. Therefore, to reduce potential differences across the treatment and control restaurants, we rely on four matching specifications to derive a sample of control restaurants that is similar to the treatment restaurants in terms of observable business characteristics. Essentially, each restaurant that receives editorial reviews is paired with a control restaurant that is not “treated” but is similar to the treatment group counterpart in terms of its probability of being treated, allowing a fair comparison of these two groups of restaurants.

1. First, for the static and categorical characteristics (i.e., food category, city, and price range), we use exact matching to pair restaurants across the treatment group, and the control group restaurants in a given

food category, city, and price range will be matched only with control group restaurants in the same food category, city, and price range.

2. Second, we drop all (145) restaurants with prior editorial reviews (i.e., restaurants with more than one editorial review by the end of our observation) as no control restaurants have prior editorial reviews. To ensure that there is no systematic difference of heterogeneity of the “prior editorial review environment” between the treatment and control groups, an exact matching must drop all treatment group restaurants with prior editorial reviews.

3. Third, we further exclude all restaurants with advertising contracts for the main analysis and use them for separate subsample and pooled sensitivity checks (see Section 6.4). This is equivalent to adding one

Figure 3. (Color online) Propensity Scores Distribution Addition of Matching for the Treatment and Control Groups

more exact matching criterion where only samples without advertising contracts exist. Such a restriction helps to ensure that there is no heterogeneity of advertising contracts and thus, no systematic difference of that heterogeneity between the treatment and control groups to prevent confounding bias introduced by the correlation between advertising contracts and editorial reviews.

4. Fourth, in addition to the aforementioned exact matchings, we apply PSM on the numerical characteristics of the matched samples from the prior three steps to enhance the validity of the matching process. Recall, however, that our data set has a unique aspect, whereby each treatment restaurant has a different treatment time (i.e., restaurants receive editorial reviews at different times), whereas none of the control restaurants have the treatment, which is not compatible with traditional PSM approaches. As such, we adopt a technique called two-stage PSM, which was used by earlier studies that faced the same challenge (Qiao et al. 2017, Yu et al. 2022). This technique also has an advantage over the conventional static PSM, as it allows dynamic matching in regard to time-varying characteristics, such as rating and review volume. In two-stage PSM, the first stage approximates restaurant i 's overall propensity score as a logistic function of a vector of all static numerical characteristics (i.e., product tags, number of available payment methods, access to free parking, and number of seats) and utilizes one-to-one nearest neighbor matching without replacements to derive the closest matched controlled and treatment restaurants pairs. This results in 1,180 pairs of restaurants. In each pair, the treatment time of the treated restaurant is taken as a “hypothetical” treatment time for the untreated counterpart. The second stage extracts the value of the time-varying characteristics, including the number of reviews and ratings at the

“hypothetical” treatment time, and calculates a logistic function to approximate restaurant i 's overall propensity score. Again, the process utilizes one-to-one nearest neighbor matching to derive matched pairs, which results in 882 pairs of restaurants. After this matching, the difference in regard to prior review environment characteristics between the treatment group restaurants and the control ones is minimized.

To evaluate the success of the PSM analysis, we compare the overall distributions of propensity scores of the matched and unmatched samples (Xu et al. 2016). Figure 3 shows the propensity score distribution addition for the treatment and control groups. It is clear that the discrepancy of the propensity score between the treatment and control units before the matching is largely erased and that the distributions of the propensity score after the matching are almost identical across the treatment units and the control ones.³ We also conduct a balance test to compare all the numerical covariates used in the matching process, as shown in Table 3. The results suggest that none of the variables are significantly different between the treatment and control groups, indicating that the control group is comparable with the treatment group before the intervention.⁴

In summary, our final data set consists of 1,764 restaurants with 86,185 reviews. We provide the complete summary statistics of all relevant variables in Table 4.

4.1.2. DID. After we constructed a comparable treatment group and control group in regard to prior review environment characteristics, advertising contracts, prior editorial review characteristics, and other business characteristics, we utilize the DID regression to uncover the causal effects of editorial reviews. In particular, our regression specification for the first

Table 3. Balance Tests on Numerical Covariates (Pretreatment) After Matching

Variables	Treatment				Control				Difference <i>p</i> -value
	Mean	Standard deviation	Min	Max	Mean	Standard deviation	Min	Max	
<i>Prior Vol</i>	14.490	24.503	0	310	19.448	24.584	0	247	0.425
<i>Prior Rate</i>	3.929	0.559	1	5	3.893	0.502	1	5	0.195
<i># of Product Tags</i>	3.561	3.815	0	31	3.527	3.996	0	35	0.790
<i># of Payment Methods</i>	2.416	0.493	1	3	2.416	0.493	1	3	0.972
<i>Free Parking (dummy)</i>	0.219	0.414	0	1	0.222	0.415	0	1	0.869
<i># of Seats</i>	22.648	14.889	2	400	22.773	17.866	2	400	0.822

part of our analysis that focuses on aggregated review characteristics is

$$Y_{it} = \beta EditorialReview_{it} + \alpha_i + \gamma_t + \zeta_{it} + \epsilon_{it}, \tag{1}$$

where Y includes *ReviewVolume*, *Rating*, and *Length* of restaurant i in month t .⁵ The independent variable of interest is the treatment indicator, *EditorialReview_{it}*, which indicates whether an editorial review is provided for restaurant i by month t .⁶ To account for the time-invariant unobserved factors that are specific to the restaurant and may affect user reviews (e.g.,

geographic location of the restaurant), we incorporate a restaurant fixed effects term α_i . To account for the time-variant unobservables that affect all businesses, we include a time fixed effects term γ_t , which captures seasonality in the change of user review characteristics. In addition, we note that review characteristics tend to change over different phases in the review life cycle, and the phases for different restaurants vary at a fixed time. We further include a relative time fixed effects term ζ_{it} to control for review phase-specific unobserved heterogeneity, where the relative time refers

Table 4. Summary Statistics

Variables	Mean	Standard deviation	Min	Max
User review characteristics				
<i>Monthly Review Volume^a</i>	1.457	1.885	1.000	532.000
<i>Monthly Rating^a</i>	3.833	0.920	1.000	5.000
<i>Rating</i>	3.863	0.876	1.000	5.000
<i>Monthly Length^a</i>	401.975	511.078	0.000	41,600.000
<i>Length</i>	502.866	605.890	0.000	42,748.000
<i>Food Course</i>	0.227	0.272	0.000	0.998
<i>Dessert & Drink</i>	0.305	0.299	0.000	0.997
<i>Emoji</i>	0.127	0.177	0.000	0.998
<i>Service</i>	0.175	0.226	0.000	0.998
<i>Atmosphere</i>	0.166	0.205	0.000	0.998
<i>Positive Score</i>	0.183	0.124	0.000	1.000
<i>Negative Score</i>	0.031	0.047	0.000	1.000
<i>Neutral Score</i>	0.787	0.123	0.000	1.000
<i>SMOG Score</i>	6.260	2.905	0.000	17.500
<i>Ad</i>	0.000	0.000	0.000	0.000
<i>Prior Vol^b</i>	43.818	55.679	1.000	427.000
<i>Prior ER Vol^b</i>	0.000	0.000	0.000	0.000
<i>Prior Rate^b</i>	3.888	0.422	1.000	5.000
<i>Experience^{a,b}</i>	10.295	12.884	0.000	79.000
Editorial review characteristics				
<i>Editorial Review (Treatment)</i>	0.484	0.634	0.000	1.000
<i>Length</i>	2,137.417	1,102.662	70.000	6,978.000
<i>Food Course</i>	0.317	0.260	0.014	0.923
<i>Dessert & Drink</i>	0.558	0.271	0.019	0.939
<i>Emoji</i>	0.034	0.022	0.013	0.347
<i>Service</i>	0.037	0.058	0.016	0.752
<i>Atmosphere</i>	0.054	0.078	0.013	0.796
<i>Positive Score</i>	0.134	0.047	0.000	0.575
<i>Negative Score</i>	0.017	0.015	0.000	0.150
<i>Neutral Score</i>	0.849	0.048	0.425	1.000
<i>SMOG Score</i>	7.788	1.092	0.000	13.400

Note. Review level or noncumulative unless otherwise noted.

^aA monthly-level variable.

^bCumulative variable.

to the time difference (by month) between the posted time of a focal review and that of the first review for the same restaurant. Finally, ϵ_{it} represents the error term. This specification allows us to estimate the effect of editorial reviews on subsequent user reviews by observing the key coefficient β . To reduce heteroscedasticity concerns, we leverage robust standard errors clustered at the restaurant level (Bertrand et al. 2004).

For the second part of the analysis, in which we focus on the textual content of the review, the unit of our analysis is review level. Therefore, our regression specification becomes

$$Y_{ikt} = \beta \text{EditorialReview}_{ikt} + \alpha_i + \gamma_t + \zeta_{it} + \epsilon_{ikt}, \quad (2)$$

where subscript ikt represents the k th user review for restaurant i in month t . The dependent variable, Y , here includes the five topics' proportions (i.e., *Food Course*, *Dessert & Drink*, *Emoji*, *Service*, and *Atmosphere*), *Positive/Negative/Neutral Score*, and *SMOG Score*. The independent variable of interest in this specification becomes $\text{EditorialReview}_{ikt}$, which indicates whether an editorial review is provided for restaurant i as of the posted time of the k th review in month t . In the same way, restaurant fixed, time fixed effects, and relative time fixed effects are included, and ϵ_{ikt} represents the error term.

4.2. Relative Time Model

The validity of our primary identification strategy that leverages the matching and DID model relies critically on the pretreatment parallel trend assumption (i.e., that there is no significant difference between treatment and control restaurants before the treatment). To test this assumption, we utilize the relative time model with the leads and lags periods (Autor 2003). With this model, we add a series of time dummies that indicate the relative chronological distance between observation time and the time one editorial review is written for restaurant i . For the first part of the analysis, in which we focus on the aggregated review characteristics, the specification of our relative time model is as follows:

$$Y_{it} = \sum_j \tau_j \text{Pre}_{it}(j) + \sum_l \omega_l \text{Post}_{it}(l) + \alpha_i + \gamma_t + \zeta_{it} + \epsilon_{it}, \quad (3)$$

where Y_{it} represents one of review characteristics (i.e., review volume, content length, and rating) for restaurant i in month t . In addition, α_i , γ_t , and ζ_{it} denote restaurant fixed effects, time fixed effects, and relative time fixed effect, respectively. The newly added term $\text{Pre}_{it}(j)$ is an indicator function that equals one if month t is j month(s) prior to the treatment. Here, we consider the posted time of each restaurant's editorial review as the treatment time. For example, if one user review was posted 0–30 day(s) earlier than the editorial review, then the indicator $\text{Pre}_{it}(1) = 1$. Similarly, the

term $\text{Post}_{it}(l)$ is an indicator function that equals one if month t is l month(s) after the editorial review is written. For simplicity, we set the indicator $\text{Post}_{it}(0) = 1$ if the user review was posted 0–30 day(s) after the editorial review. To estimate all of the effects, we gather all pretreatment periods that are greater than or equal to six months prior to treatment into one dummy. Similarly, we assemble all posttreatment periods that are greater than or equal to six months following treatment into another dummy. Hence, the coefficient τ_j for $j = \geq -6, -5, \dots -1$ captures the pretreatment trend of the impact of editorial reviews on user reviews, whereas the coefficient ω_l for $l = 0, 1, 2, \dots 5, \geq 6$ captures the effect of editorial reviews in each posttreatment period. Consistent with prior work, we set a period prior to the time of treatment as the baseline by normalizing the coefficient of $\text{Pre}_{it}(-1)$ to zero.

As for the second part of the analysis at the review level, the specification of our relative time model is

$$Y_{ikt} = \sum_j \tau_j \text{Pre}_{ikt}(j) + \sum_l \omega_l \text{Post}_{ikt}(l) + \alpha_i + \gamma_t + \zeta_{it} + \epsilon_{ikt}. \quad (4)$$

As before, Y_{ikt} represents one of the textual characteristics of the reviews (i.e., topics, sentiment, and readability) for restaurant i the posted time of the k th review in month t . The term $\text{Pre}_{ikt}(j)$ is an indicator function that equals one if the posted time of the k th review in month t is j month(s) prior to the treatment, and similarly, we consider the posted time of each business editorial review as the treatment time. That is, if one user review was posted 0–30 day(s) earlier than the first editorial review, then the indicator $\text{Pre}_{ikt}(1) = 1$. Consistently, the term $\text{Post}_{ikt}(l)$ is an indicator function that equals one if the posted time of the k th review in month t is l month(s) after the editorial review is written. That is, $\text{Post}_{ikt}(0) = 1$ if the user review was posted 0–30 day(s) after the editorial review. α_i , γ_t , and ζ_{it} denote restaurant fixed effect, time fixed effect, and relative time fixed effect, respectively.

4.3. Mechanism Analysis

In this subsection, we present additional analyses to understand further the underlying mechanisms for the change of the reviewers' review-writing behaviors caused by editorial reviews. The purpose of our mechanism analysis is to supplement our main analyses, which focus on whether and to what degree editorial reviews causally influence subsequent user reviews with respect to quantitative- and qualitative-related characteristics. In this regard, we seek to examine how and why the subsequent reviews are changed. Recall that one of the theoretical constructs that is usually used in the literature in the context of online reviews is herding and/or differentiation. In particular, prior

works have consistently demonstrated that the review generation, rating, and writing behaviors of subsequent reviewers are all influenced by their earlier peers (Schlosser 2005, Moe and Trusov 2011, Moe and Schweidel 2012, Lee et al. 2015). Because editorial reviews are presented in a similar manner to those generated by consumers in the context of this study, we expect that a similar influence may exist between editors and subsequent noneditor users. Specifically, on the one hand, the editorial reviews might trigger more users to write reviews (i.e., herding in regard to choosing which restaurant to visit and subsequently, to write a review for) or to follow the editorial reviews in certain dimensions (e.g., topics, sentiment, length, readability) of composing reviews (i.e., herding in regard to how to write reviews). On the other hand, users may be discouraged to review for the restaurant reviewed by editorial reviews (i.e., differentiation regarding choosing which restaurant to visit and subsequently, to write a review for) or to review in a different way from the editorial reviews (i.e., differentiation regarding how to write reviews). At the review level, these mechanisms, if existing, should collectively result in differences in the similarity between editorial reviews and user reviews before and after editorial reviews.

Note that our main analyses already reveal the underlying mechanisms, partially through the observed changes in review volume. Based on the DID specification, an increase/decrease in review volume captures the change in the decision of the same set of platform users on whether to visit and subsequently, to write a review for restaurants that receive editorial reviews. Such a change can be interpreted as herding/differentiation of users toward editorial reviews in regard to choosing which restaurants to visit and therefore, to write a review for.⁷ However, the mechanism that determines how users write reviews remains unclear. To further uncover this perspective, we conduct additional empirical analyses at two levels.

1. First, we investigate the mechanism at the review level to attribute the effect of editorial reviews found in the main analysis to whether the subsequent reviews become more similar or dissimilar to editorial reviews. Despite not disentangling herding/differentiation in regard to how to write a review from other mechanisms/factors (e.g., endogenous group forming) (Moffitt et al. 2001), the answer is valuable from a normative perspective, as it informs practitioners as to what the subsequent user reviews are like after posting an editorial review (e.g., more similar or dissimilar to the editorial review) and how to write editorial reviews for desired subsequent user reviews.

2. Second, we further consider the reviewer-level behavioral change triggered by the program and isolate herding/differentiation in regard to how to write a review from the other contextual factors and the

endogenous group formation. For instance, this analysis examines whether a group of systematically different reviewers, who inherently write reviews that are similar or dissimilar to the editorial reviews, is more inclined to provide reviews for the restaurants with editorial reviews. If a user's reviews become more similar to editorial reviews after the user's exposure to the editorial review, we attribute that to herding behavior. In contrast, if a user's reviews become more polarized after exposure to editorial reviews, then such a change can be attributed to differentiation behavior. As compared with the review-level analysis, this analysis is more important from a theoretical perspective, as it attributes the impact of editorial reviews to individuals' herding/differentiation behaviors.

For both levels of analyses, we need to first measure the similarity between editorial reviews and user reviews. Specifically, we first quantify the characteristics of editorial reviews in a manner consistent with those of user reviews using the methods noted in Section 3.2.2. Using the extracted topic distributions, sentiment scores, and readability score for editorial reviews, we then construct (1) topic similarity, (2) sentiment/rating similarity, (3) length similarity, and (4) readability similarity to measure the content similarity in all review-level characteristics, as described.

4.3.1. Topic Similarity. Because each review is represented as a topic vector, we measure the topic similarity between editorial reviews and user reviews using cosine similarity. Cosine similarity assesses the cosine of the angle between two nonzero vectors. The cosine similarity between two topic vectors reflects how similar these reviews are in terms of covered topics. Specifically, if the cosine similarity equals one, it means that these two reviews are identical in regard to the topic distribution. In contrast, if the cosine similarity equals zero, it means that these two reviews are completely different from each other with respect to topics. In our analysis, for each user review, we measure the topic similarity between that user review and the editorial review. For notation, we denote the topic similarity between the k th user review for business i in month t and the editorial review for business i as $\Delta Topic_{ikt}$. Here, T_{ikt}^{UR} and T_i^{ER} are the topic vectors of the user review and the editorial review, respectively. Thus, the cosine similarity between topic vectors of editorial reviews and user reviews is as follows:

$$\Delta Topic_{ikt} = \text{Cos}(T_{ikt}^{UR}, T_i^{ER}). \quad (5)$$

4.3.2. Sentiment/Rating Similarity. We evaluate the sentiment similarity between editorial reviews and user reviews in a manner that resembles how we measure topic similarity. In particular, we compare the user review with editorial reviews. Instead of using

cosine similarity, we focus on the absolute difference of each sentiment score. Hence, we specify the sentiment similarity as follows:

$$\Delta \text{Sentiment}_{ikt} = |S_{ikt}^{UR} - S_i^{ER}|, \quad (6)$$

where $\Delta \text{Sentiment}_{ikt}$ represents the difference of the sentiment scores between the k th user review in month t for business i and the editorial review for business i . Note that we calculate positive score S^{pos} , neutral score S^{neu} , and negative score S^{neg} to represent the sentiment of each review body. S_{ikt}^{UR} is a vector of three values (i.e., $[S_{ikt}^{UR,pos}, S_{ikt}^{UR,neu}, S_{ikt}^{UR,neg}]$), representing the sentiment scores of the k th user review for business i at time t . Consistently, $\Delta \text{Sentiment}_{ikt}$ and S_i^{ER} are vectors of three elements.

Note that the editorial reviews do not have an explicit rating. Research has established that rating reflects sentiment and the consistency between review (textual) sentiment and rating (Geetha et al. 2017), and our analysis on the sentiment shows that the “treatment” user reviews tend to be more neutral and less positive because of herding to the editorial reviews, which aggregately suggests lower sentiment and can explain the declines of the overall rating. We thus construct $\Delta \text{Rating}_{ikt}$ as follows:

$$\Delta \text{Rating}_{ikt} = |\text{Rating}_{ikt}^{UR} - \text{RatingEst}_i^{ER}|, \quad (7)$$

where $\Delta \text{Rating}_{ikt}$ is the similarity (measured as the absolute difference) between the rating of the k th user review posted for business i in month t , Rating_{ikt}^{UR} , and an estimated rating of the editorial review for business i , RatingEst_i^{ER} , inferred from its sentiment in the review text. To construct the latter, we regress all of the instances of Rating_{ikt}^{UR} on their sentiment variables $S_{ikt}^{UR,pos}$ and $S_{ikt}^{UR,neu}$ to extract the correlation between ratings and sentiments.⁸ Then, we feed the trained model with the $S_i^{ER,pos}$ and $S_i^{ER,neu}$ to obtain RatingEst_i^{ER} . A smaller $\Delta \text{Rating}_{ikt}$ indicates that the user reviews are more similar in rating and sentiment to the editorial review for the same restaurant.

4.3.3. Length Similarity. Similar to the sentiment similarity and rating similarity, the length similarity between editorial reviews and user reviews is represented by

$$\Delta \text{Length}_{ikt} = |\text{ContentLength}_{ikt}^{UR} - \text{ContentLength}_i^{ER}|, \quad (8)$$

where $\Delta \text{Length}_{ikt}$ is the content length similarity, measured as the absolute value of the difference between the content length of k th user review posted for business i in month t , $\text{ContentLength}_{ikt}^{UR}$, and that of the editorial review, $\text{ContentLength}_i^{ER}$. A smaller $\Delta \text{Length}_{ikt}$ indicates that the user review is more similar to the editorial review regarding the content length.

4.3.4. Readability Similarity. Similar to the specification, the readability similarity between editorial reviews and user reviews can be represented by

$$\Delta \text{SMOG}_{ikt} = |\text{SMOG}_{ikt}^{UR} - \text{SMOG}_i^{ER}|, \quad (9)$$

where ΔSMOG_{ikt} is the readability similarity between the k th user review posted for business i in month t and the editorial review for business i ; SMOG_{ikt}^{UR} is the SMOG score of the k th user review posted for business i in month t ; and SMOG_i^{ER} is the SMOG score of the editorial review for business i .

To reveal the review-level mechanism, we estimate the following equation:

$$\begin{aligned} \text{Similarity}_{ikt} = & \beta \text{EditorialReview}_{ikt} + \theta_1 \text{PriorVol}_{ikt} \\ & + \theta_2 \text{PriorRate}_{ikt} + \alpha_i + \gamma_t + \zeta_{it} + \epsilon_{ikt}, \end{aligned} \quad (10)$$

where subscript ikt represents the k th user review for restaurant i in month t . The dependent variable Similarity_{ikt} refers to (an element of) one of the following: ΔTopic , $\Delta \text{Sentiment}$, ΔRating , ΔLength , and ΔSMOG . As before, $\text{EditorialReview}_{ikt}$ concerns whether an editorial review for restaurant i is provided as of the posted time of the k th review in month t . PriorVol_{ikt} and PriorRate_{ikt} are the prior review volume and cumulative rating, respectively, for restaurant i by the posted time of the k th review in month t , which we add to control for user review environment changes.⁹ α_i , γ_t , and ζ_{it} denote restaurant fixed effect, time fixed effect, and relative time fixed effect, respectively. Using the similarity measure as the dependent variable, we show whether the content of user reviews that are posted after the editorial reviews is more similar to that of the editorial reviews than their before editorial review counterparts. Note that, for this part of the study, we utilize only a subset of data (41,491 observations), which includes only treatment restaurants because we are interested in the reviews pertaining only to such restaurants. The reviews of the nontreatment restaurants are not affected by any editorial review at any time period and thus, do not have a paired editorial review to measure their degree of similarity. In addition, before feeding the data into the model, we follow Hosanagar et al. (2014) to use a bootstrapping method to achieve volume equalization. Because there are 28,497 after-treatment user reviews, we bootstrapped 28,497 before-treatment reviews. Such equalization rules out the alternative explanation of review volume difference for the changes in content similarity.¹⁰

The reviewer-level analysis follows the same process as the review-level analysis explained. We narrow down the analysis scope to long-time users by keeping only those reviews by reviewers who post reviews both before and after the presence of editorial reviews. This is because of the inherent difficulty in

teasing out the effect of endogenous group forming (i.e., homophily) in cross-sectional data of individuals (Manski 1993, Moffitt et al. 2001). The herding/differentiation behavior of the short-time reviewers is, thus, unidentifiable. After filtering out the recent users, we have repeated observations for each reviewer r . Using these data,¹¹ following the identification strategy of Nair et al. (2010), we can tease out endogenous group forming through the regression as follows:

$$\begin{aligned} \text{Similarity}_{irkt} = & \beta \text{EditorialReview}_{irkt} + \theta_1 \text{PriorVol}_{ikt} \\ & + \theta_2 \text{PriorRate}_{ikt} + \theta_3 \text{Experience}_{rt} + \alpha_i \\ & + \tau_r + \gamma_t + \zeta_{it} + \epsilon_{irkt}, \end{aligned} \quad (11)$$

where subscript $irkt$ represents the k th user review for restaurant i in month t generated by reviewer r . Hence, $\text{EditorialReview}_{irkt}$ concerns whether an editorial review for restaurant i is provided as of the posted time of the k th review in month t generated by reviewer r , and ϵ_{irkt} is the unexplained error term. τ_r is a reviewer fixed effect that controls for users' inherent and persistent review-writing similarity to editorial reviews. To allow users' inherent review-writing similarity to evolve with their experience, we also control for the experience of reviewer r at time t , Experience_{rt} . All other specifications are the same as in Equation (10). Accordingly, we use only a within-prior user review environment (rating and volume), within-reviewer, within-experience, within-time, within-restaurant, and across-editorial-review variation in content similarity to identify reviewers' behavioral change.

It is worth noting that our mechanism analyses rely on an assumption that contextual effects and inherently changing taste/behaviors of individuals are reasonably controlled for. In other words, it is important to control for other exogenous factors and inherent changes in review-writing similarity to editorial reviews that are not because of the editorial review (Manski 1993, Moffitt et al. 2001, Nair et al. 2010, Lee et al. 2015). Specifically, the assumptions are that Experience_{rt} reasonably controls for such inherently changing taste/behaviors and that the other controlling specification (i.e., PriorVol_{ikt} , PriorRate_{ikt} , α_i , γ_t , and ζ_{it}) reasonably controls for contextual effects. As such, the estimate of β is theoretically a close and upper bound for the behavioral change of review writing, influenced by editorial reviews (Nair et al. 2010).

5. Empirical Results

5.1. Relative Time Model

We first report the results of the relative time model to establish that the parallel trend assumption holds in our analysis so our matching + DID specification is valid. As shown in Table 5, we find none of the coefficients of the pretreatment dummies, $\text{Pre}(j)$, to be statistically significant at a p -value of < 0.10 . This result

confirms that there is no detectable pretreatment dissimilarity across reviews for restaurants that receive editorial reviews or for those that do not receive editorial reviews after the matching process; hence, the parallel trend assumption is satisfied.

5.2. Aggregated Review Characteristics

The estimation results for the impact of editorial reviews on aggregated characteristics of user reviews, which are performed at the restaurant-monthly level, are shown in Table 6. First, we find a strong and significant increase in the number of new user reviews (i.e., ReviewVolume) after the presence of editorial reviews. From an economic perspective, this translates to an increase of 3.1% user reviews per month for restaurants with one editorial review. The discussion of underlying mechanisms related to review volume (see Section 4.3) implies that platform users tend to write more reviews for restaurants that receive editorial reviews, suggesting the herding of users toward editors in regard to which restaurants to review. A possible next-layer mechanism, as shown in the literature, is that the posting of editorial reviews can increase consumers' awareness of the restaurant (Berger et al. 2010). Hence, after receiving editorial reviews, the restaurants may attract more attention from consumers (i.e., attain more consumers) and hence, receive more reviews.¹² In addition, the content of editorial reviews may trigger users' willingness to express their opinions, especially when they are inspired by agreed upon but not previously discussed topics or want to share their disagreement with the content of editorial reviews.

Similarly, the existence of editorial reviews increases the *Length* of the subsequent user reviews for the same restaurants. All else being equal, this finding suggests that user reviews written for restaurants with an editorial review are about 3.6% longer, on average, than are those written for restaurants without editorial reviews. A plausible interpretation is that editorial reviews serve as a starting point for discussions. The comprehensive content of editorial reviews may inspire reviewers to express their opinions about restaurants in a more thorough fashion. For instance, reviewers may provide only a simple evaluation of the food (e.g., delicious or not) before the existence of editorial reviews. After reading editorial reviews, however, reviewers may begin to assess the food from various perspectives, such as flavor, smell, appearance, and so forth. We will explore this possibility by examining the content of user reviews influenced by editorial reviews in Sections 5.3 and 5.4.

Further, we find a negative effect of editorial reviews on the dependent variable *Rating*, although the statistical power of such an effect is rather weak ($p < 0.10$). This indicates that the review valence tends to

Table 5. Results from the Relative Time Model

DV	log(Review Volume)	Rating	log(Length)	Food Course	Dessert & Drink	Emoji	Service	Atmosphere	Positive	Neutral	Negative	SMOG
<i>Pre</i> (≥ -6)	-0.020 (0.013)	-0.007 (0.037)	0.036 (0.055)	0.002 (0.007)	-0.010 (0.006)	0.008 (0.006)	0.002 (0.008)	0.006 (0.007)	0.002 (0.004)	0.002 (0.004)	0.000 (0.002)	-0.072 (0.101)
<i>Pre</i> (-5)	-0.021 (0.17)	-0.012 (0.036)	0.033 (0.041)	0.004 (0.011)	-0.004 (0.012)	0.005 (0.008)	0.005 (0.011)	0.008 (0.010)	0.002 (0.006)	0.002 (0.006)	0.001 (0.002)	-0.210 (0.145)
<i>Pre</i> (-4)	0.019 (0.015)	-0.022 (0.033)	0.011 (0.038)	0.009 (0.011)	-0.010 (0.012)	0.003 (0.008)	0.003 (0.011)	0.001 (0.010)	0.000 (0.006)	0.005 (0.006)	-0.000 (0.002)	-0.213 (0.144)
<i>Pre</i> (-3)	0.004 (0.014)	-0.023 (0.030)	0.023 (0.035)	0.016 (0.010)	-0.015 (0.012)	0.000 (0.008)	0.015 (0.010)	0.007 (0.009)	-0.004 (0.006)	0.006 (0.006)	0.001 (0.002)	-0.078 (0.139)
<i>Pre</i> (-2)	0.010 (0.013)	-0.035 (0.027)	0.022 (0.032)	0.014 (0.009)	0.002 (0.011)	0.002 (0.007)	-0.005 (0.010)	0.007 (0.009)	-0.008 (0.006)	0.007 (0.006)	0.003 (0.002)	-0.071 (0.129)
<i>Pre</i> (-1)												
Baseline												
<i>Post</i> (0)	0.056*** (0.011)	-0.019 (0.023)	0.070** (0.028)	0.005 (0.009)	0.005 (0.010)	0.002 (0.007)	-0.007 (0.009)	-0.001 (0.008)	-0.011** (0.005)	0.011** (0.005)	0.003 (0.002)	-0.042 (0.118)
<i>Post</i> (1)	0.042*** (0.011)	-0.021 (0.023)	0.049* (0.028)	0.009 (0.009)	0.010 (0.010)	0.000 (0.007)	-0.011* (0.009)	-0.015* (0.008)	-0.012** (0.005)	0.012** (0.005)	0.003 (0.002)	0.037 (0.124)
<i>Post</i> (2)	0.032*** (0.012)	-0.023 (0.024)	0.049* (0.029)	0.013 (0.009)	0.020** (0.010)	-0.004 (0.007)	-0.016* (0.009)	-0.020** (0.008)	-0.013*** (0.005)	0.016*** (0.005)	0.002 (0.002)	0.090 (0.126)
<i>Post</i> (3)	0.023* (0.012)	-0.042* (0.024)	0.048* (0.029)	0.016* (0.009)	0.021** (0.010)	-0.009 (0.007)	-0.022** (0.009)	-0.021** (0.008)	-0.012** (0.005)	0.011** (0.005)	0.003 (0.002)	0.116 (0.125)
<i>Post</i> (4)	0.022* (0.012)	-0.056** (0.025)	0.050* (0.030)	0.018** (0.009)	0.027** (0.010)	-0.012* (0.007)	-0.028*** (0.010)	-0.027*** (0.009)	-0.011** (0.006)	0.009 (0.006)	0.002 (0.002)	0.232* (0.128)
<i>Post</i> (5)	0.021* (0.012)	-0.044* (0.026)	0.066** (0.031)	0.020** (0.009)	0.032*** (0.010)	-0.013* (0.007)	-0.016* (0.010)	-0.027*** (0.009)	-0.010 (0.006)	0.010 (0.006)	0.001 (0.002)	0.111 (0.130)
<i>Post</i> (≥ 6)	0.033*** (0.09)	-0.055** (0.027)	0.031 (0.022)	0.025*** (0.007)	0.033*** (0.008)	-0.017*** (0.005)	-0.010** (0.007)	-0.016** (0.006)	-0.006 (0.004)	0.008* (0.004)	0.000 (0.002)	0.089 (0.094)
Restaurant FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Relative time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	48,027	48,027	48,027	86,185	86,185	86,185	86,185	86,185	86,185	86,185	86,185	86,185
Adjusted R^2	0.293	0.137	0.157	0.436	0.436	0.290	0.166	0.200	0.096	0.091	0.059	0.081

Notes. Robust standard errors are in parentheses. DV, dependent variable; FE, fixed effect.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 6. Effect of Editorial Reviews on Aggregated Review Characteristics

DV	log(Review Volume)	Rating	log(Length)
<i>Editorial Review</i>	0.031*** (0.006)	−0.022* (0.013)	0.035** (0.014)
Restaurant FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Relative time FE	Yes	Yes	Yes
Observations	48,027	48,027	48,027
Adjusted R^2	0.318	0.137	0.163

Notes. Robust standard errors are in parentheses. DV, dependent variable; FE, fixed effect.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

decrease for restaurants that receive editorial reviews. At first glance, this result may appear to be surprising given that editorial reviews are often perceived to be similar to advertorial ones. Thus, one may expect that editorial reviews may positively influence the ratings of subsequent reviews. However, it turns out that editorial reviews, which are designed to be neutral, appear to drive subsequent user reviews to have lower average star ratings. We examine the change in reviewers' behavior following editorial reviews further in Sections 5.3 and 5.4.

According to the results, it appears that editorial reviews stimulate reviewers not only to contribute more reviews to the platform but also, to spend more effort in doing so, which results in greater length of content. From this perspective, editorial reviews can be considered a valuable addition that could stimulate greater user contribution to the overall review platform.

5.3. Characteristics of Textual Content

Next, we examine the changes in the textual content of user reviews that are influenced by the presence of editorial reviews. The analysis in this section helps us to understand more about the change in review writing that follows editorial reviews. Table 7 presents the results from such an analysis, which is performed at the review level.

Recall that *Dessert & Drink* and *Food Course* occupy the two largest portions of the editorial review content (see Table 4) and that the proportion of these topics is larger in editorial reviews than in user reviews (i.e., editorial reviews more often discuss food and drink). The positive and statistically significant coefficients of *EditorialReview* for *Dessert & Drink* and *Food Course* indicate that the presence of editorial reviews influences user reviews to more often discuss food and drink (i.e., *Dessert & Drink* and *Food Course*) as well. At the same time, the proportion of *Emoji*, *Service*, and *Atmosphere* shrinks. In short, the content structure of user reviews gradually resembles that of editorial reviews.

It is worth noting, however, that a decrease in topic proportion does not mean a decrease in the content length for that topic because we normalize the content length to calculate the proportion. In Section 6.1, we consider length and topic proportion jointly to examine the net changes in topic content length and thus, content variety.

In addition, the presence of editorial reviews greatly affects the sentiment of subsequent user reviews. Specifically, the treatment effect on neutral scores of user reviews is significantly positive at $p < 0.05$. This finding indicates that reviews become more neutral if they are contributed after the editorial reviews are posted for the focal restaurant. At the same time, the positive sentiment of user reviews decreases at $p < 0.05$. These results are consistent with the finding in the previous subsection, which suggests that user reviews become more neutral (i.e., the aggregated review valence decreases), and provide more detail about the change in the reviewers' behavior in regard to the review sentiment. Note that the neutral score of editorial reviews is 0.85, on average, and is higher than that of user reviews in our data set. Following the neutral perspectives expressed in the editorial reviews, reviewers appear to be more objective when they evaluate restaurants, resulting in a slight decline of overall review valence. Finally, the readability of user reviews does not experience a significant change after editorial reviews are posted.

5.4. Underlying Mechanisms

In this section, we report results from our mechanism analysis to further our understanding regarding the underlying mechanisms of the changes in reviewer behavior observed in the previous subsections. Recall that each treated restaurant in our data set may receive more than one editorial review. In this section, we explore the changes in user reviews with respect to their closest editorial review. The estimation results for the review-level analysis are provided in Table 8, which shows that the content of user reviews becomes more similar to that of editorial reviews in all aspects, including content topics, sentiment (rating), length, and readability. In particular, recall that the larger value of cosine similarity measure indicates a higher degree of similarity. The significantly positive coefficient for $\Delta Topic$ indicates that the topic distribution covered by reviews published after editorial reviews is becoming more similar to the editorial review content. Specifically, reviewers tend to focus more on topics such as desserts, drinks, and main courses, which are in line with the proportion of those topics in editorial reviews.

Further, recall that the similarity measures for sentiment, rating, length, and readability are constructed as an absolute value of difference.¹³ As such, a smaller

Table 7. Effect of Editorial Reviews on Textual Review Characteristics

DV	Food Course	Dessert & Drink	Emoji	Service	Atmosphere	Positive	Neutral	Negative	SMOG
<i>Editorial Review</i>	0.009*** (0.003)	0.032*** (0.003)	−0.007*** (0.002)	−0.011*** (0.003)	−0.023*** (0.003)	−0.002** (0.001)	0.002** (0.001)	<0.001 (<0.001)	0.028 (0.039)
Restaurant FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Relative Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	86,185	86,185	86,185	86,185	86,185	86,185	86,185	86,185	86,185
Adjusted R^2	0.436	0.368	0.290	0.166	0.200	0.059	0.054	0.021	0.090

Notes. Robust standard errors are in parentheses. DV, dependent variable; FE, fixed effect.

** $p < 0.05$; *** $p < 0.01$.

value represents a larger degree of similarity. The negative and statistically significant coefficients of $\Delta Positive$, $\Delta Neutral$, and $\Delta Rating$ indicate that the sentiment (rating) of user reviews gravitates toward the sentiment of editorial reviews written for the same restaurant. Similarly, the coefficient for $\Delta Length$ is also negative and statistically significant, suggesting that the length of subsequent user reviews, as compared with those of prior user reviews, is closer to those of editorial reviews. Interestingly, the coefficient for $\Delta SMOG$ is also significantly negative, indicating that the readability of editorial and user reviews is also becoming more similar. Together with the insignificant impact found in our main results from the DID specification (Section 5.3 and Table 7), this indicates a potential “spillover effect” of editorial reviews to the readability of reviews of the nontreatment restaurants. In other words, an editorial review might influence the readability of not only the subsequent user reviews of the same restaurant but also, those of nontreatment restaurants. Such a spillover effect is also theoretically possible because readability is likely to be reviewer dependent rather than restaurant dependent. Therefore, a change in review-writing behavior in regard to review readability may not be exclusively applied to the reviews of the treatment restaurants alone.

In summary, we find strong evidence that restaurant reviews are starting to resemble editorial reviews after the editorial reviews’ publication from the perspective of topics, sentiment, and readability. This suggests that review platforms can set “exemplar” editorial reviews to derive desired subsequent user reviews that resemble the example. Interestingly, in a supplemental analysis for a different purpose, we show insignificant resembling tendencies toward top-placed noneditor user reviews in all dimensions (see Online Appendix C.3). This indicates that the effects of editorial reviews should be more than the salience effects of top-placed reviews, suggesting the existence of alternative mechanisms (e.g., herding) that are specific to editorial reviews.

The review-level analysis results provide support for herding behavior from consumers toward editors. We further validate these results by conducting additional analysis at the reviewer level using long-term reviewers. The results are shown in Table 9. The *EditorialReview* coefficients of all similarity measures are qualitatively consistent with those at the review-level analysis. Note that, in this analysis, the coefficients are identified from the within-reviewer variations. The coefficients thus are interpreted as that for a given user; his or her reviews after the editorial reviews for the same restaurants are more similar to the editorial

Table 8. Estimation Results of Review-Level Mechanism Analysis

DV	$\Delta Topic$	$\Delta Positive$	$\Delta Neutral$	$\Delta Negative$	$\Delta SMOG$	$\Delta Length$	$\Delta Rating$
<i>Editorial Review</i>	0.012*** (0.002)	−0.005*** (0.001)	−0.004*** (0.001)	0.001 (0.001)	−0.116*** (0.033)	−0.026*** (0.008)	−0.043*** (0.011)
<i>Prior Vol</i>	0.000*** (0.000)	−0.000*** (0.000)	−0.000 (0.000)	0.000 (0.000)	−0.000 (0.000)	0.000 (0.000)	0.001*** (0.000)
<i>Prior Rate</i>	−0.002 (0.003)	0.001 (0.002)	0.002 (0.002)	−0.001 (0.001)	0.060 (0.043)	0.012 (0.011)	−0.145*** (0.015)
Restaurant FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Relative time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	56,994	56,994	56,994	56,994	56,994	56,994	56,994
Adjusted R^2	0.205	0.165	0.165	0.071	0.189	0.618	0.143

Notes. Robust standard errors are in parentheses. DV, dependent variable; FE, fixed effect.

*** $p < 0.01$.

Table 9. Estimation Results of Reviewer-Level Mechanism Analysis

DV	$\Delta Topic$	$\Delta Positive$	$\Delta Neutral$	$\Delta Negative$	$\Delta SMOG$	$\Delta Length$	$\Delta Rating$
<i>Editorial Review</i>	0.015*** (0.003)	−0.004** (0.002)	−0.006*** (0.002)	−0.001 (0.001)	−0.198*** (0.045)	−0.036*** (0.014)	−0.033*** (0.013)
<i>Experience</i>	0.008*** (0.001)	−0.002* (0.001)	−0.001* (0.001)	0.000 (0.000)	−0.030** (0.013)	−0.025*** (0.004)	−0.009** (0.004)
<i>Prior Vol</i>	−0.000*** (0.000)	−0.000*** (0.000)	−0.000*** (0.000)	−0.000 (0.000)	0.000 (0.001)	0.000 (0.000)	0.001** (0.000)
<i>Prior Rate</i>	−0.012*** (0.004)	0.008*** (0.003)	0.006** (0.003)	0.003*** (0.001)	−0.293*** (0.064)	0.164*** (0.019)	−0.200*** (0.018)
Restaurant FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Reviewer FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Relative time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	51,372	51,372	51,372	51,372	51,372	51,372	51,372
Adjusted R^2	0.925	0.907	0.911	0.895	0.908	0.959	0.945

Notes. Robust standard errors are in parentheses. DV, dependent variable; FE, fixed effect.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

reviews than were his or her reviews before, which aligns with the herding behavior of users in their review generation. This suggests that editorial reviews may exhibit gravitational powers that persuade reviewers to follow while contributing their own content. Therefore, by providing high-quality editorial reviews, the review platform can nudge reviewers to provide longer and richer feedback. Taken together with the finding of increased review volume triggered by editorial reviews, our analysis suggests the herding of users toward platform editors in terms of choosing which restaurants to review and how to write reviews.

6. Extended Analysis and Robustness Checks

6.1. Content Length for Topics

In the main analysis, we consider the impact of editorial reviews on the proportion of each topic in subsequent user reviews. The results may, however, produce misleading interpretations about topic diversity. The decrease in the proportion of *Emoji*, *Service*, and *Atmosphere* does not necessarily indicate decreased content length of such topics and thus, decreased topic diversity after considering that the total content length increases, as shown

in Section 5.2. To reveal the impact of editorial reviews on topic diversity, we investigate the impact of editorial reviews on the content length of each topic (i.e., *Topic Length*) of the subsequent user reviews. To calculate *TopicLength*, we multiply the review-level *ContentLength* with the relative topic proportion (i.e., *FoodCourse*, *Dessert & Drink*, *Emoji*, *Service*, and *Atmosphere*). This is in line with LDA topic modeling that extracts the relative proportion of each topic as the topic length to content length ratio. We then estimate the impact of editorial reviews on *TopicLength* using the main DID specification in Equation (2), with dependent variable $\log(\text{TopicLength})$. As shown in Table 10, we find that the topic lengths for *Food Course* and *Dessert & Drink* significantly increase, whereas the topic lengths for the other topics do not change significantly. This suggests that, although the editorial reviews increase the content on *Food Course* and *Dessert & Drink*, they do not decrease the content on other topics. This indicates a net increase in content variety contributed by the increased length for *Food Course* and *Dessert & Drink*.

6.2. Alternative Specifications for Ratings, Length, and Review Volume

For aggregate review characteristics (i.e., *Rating*, *Length*, and *ReviewVolume*), it is possible that editorial reviews

Table 10. Effect of Editorial Reviews on Topic Length

DV	$\log(\text{Food Course Length})$	$\log(\text{Dessert \& Drink Length})$	$\log(\text{Emoji Length})$	$\log(\text{Service Length})$	$\log(\text{Atmosphere Length})$
<i>Editorial Review</i>	0.066*** (0.025)	0.146** (0.024)	−0.032 (0.023)	−0.034 (0.024)	−0.035 (0.023)
Restaurant FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Relative time FE	Yes	Yes	Yes	Yes	Yes
Observations	86,185	86,185	86,185	86,185	86,185
Adjusted R^2	0.323	0.255	0.109	0.204	0.135

Notes. Robust standard errors are in parentheses. DV, dependent variable; FE, fixed effect.

** $p < 0.05$; *** $p < 0.01$.

Table 11. Alternative Specifications for Ratings, Length, and Review Volume

DV	Restaurant-month level			Review level		Negative binomial	
	log(Review Volume)	Rating	log(Length)	Rating	log(Length)	Review Volume	Length
<i>Editorial Review</i>				−0.020* (0.012)	0.027** (0.012)	0.026** (0.013)	0.023** (0.008)
<i>Lagged Editorial Review</i>	0.020*** (0.006)	−0.021* (0.012)	0.030** (0.014)				
Restaurant FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Relative time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	48,027	48,027	48,027	86,185	86,185	48,027	48,027
Adjusted/pseudo- R^2	0.318	0.137	0.163	0.113	0.123	0.152	0.119

Notes. Robust standard errors are in parentheses. DV, dependent variable; FE, fixed effect.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

for restaurants are provided in the middle of time T . Our current specification thus results in an underestimated treatment effect. Note that we also have review-level observations for *Rating* and *Length*, and the review-level analysis that uses Equation (2) does not suffer from this treatment effect. Hence, we conduct an additional review-level analysis for *Rating* and *Length*. We further alternatively specify the treatment in Equation (1) as $LaggedEditorialReview_{it}$, in which if restaurant i is treated in the middle of T , $LaggedEditorialReview_{it} = 0$, whereas $LaggedEditorialReview_{it} = 1$ as t starts from $T + 1$. The results of both alternative specifications, shown in Table 11, are consistent with the main results in Table 6. In addition, note that both *Length* and *ReviewVolume* are count variables, and we consider the alternative model framework of negative binomial regression. The results, also shown in Table 11, are consistent with the main results.

6.3. Controlling for Prior Review Environment in Regression

In our main analysis, we control for prior review environment (e.g., rating and review volume) in the matching process. An alternative approach to control for such factors is to add *PriorVol* and *PriorRatings* as control variables in the DID regressions in Equations (1) and (2).¹⁴ Note that we do not add prior editorial review characteristics because all of the data points can have, at most, one editorial review and no prior ones. The results, shown in Tables 1 and 2 in Online Appendix B, are consistent with the main analysis results.

We note that, despite consistency, the estimated effects are theoretically biased from the causal effects of editorial reviews. Controlling for the prior user reviews blocks any indirect causal effects of editorial reviews on focal user reviews through other user reviews in between. Consequently, the estimated treatment effects are closer to the direct causal effects. To correct for this bias, for all of the user reviews of the controlled restaurants and of the treatment restaurants before their treatment, we keep the same specification,

whereas for the “treated” user reviews, we set *PriorVol* and *PriorRatings* to be the values at the treatment time. This improved specification prevents blocking of the noted indirect causal effects. The estimation results, shown in Tables 3 and 4 in Online Appendix B, are again consistent with our main results.¹⁵

6.4. Additional Analysis for Restaurants with Advertising Contracts

Our main analysis excludes the restaurants with advertising contracts to ensure that our results are not confounded by these contracts. To demonstrate the generalizability of our results, we conduct two additional analyses using those excluded restaurants. The first analysis uses the restaurants with advertising contracts only, and the second pools such restaurants with the restaurants for the main analysis (i.e., restaurants never with advertising contracts). The general empirical framework is the same as our main analysis, except that we include the indicator of whether restaurant i at time t has an advertising contract Ad_{it} as an additional covariate in the second stage of the two-stage PSM and as a control variable in the mechanism analysis. As shown in Tables 5–12 in Online Appendix B, the impact of editorial reviews remains similar to the main analysis and the mechanism analysis at all levels.

6.5. Including Heterogeneous Treatment Effects of Sentiment

In this section, we consider the heterogeneity effects of editorial reviews. Note that restaurants in our main analysis all have the same editorial review volume of one. We thus focus on the heterogeneity in the sentiment of these editorial reviews. Specifically, we add demeaned *NegativeScore* and *NeutralScore* of the treatment editorial reviews¹⁶ and interact them with the treatment. The results are shown in Tables 13 and 14 in Online Appendix B. For mechanism analysis, we also include these two variables to examine their varying impact. The results are shown in Tables 15 and 16

in Online Appendix B. The effects of *EditorialReview* (i.e., the average treatment effect in the main specification) are consistent with our main results. Interestingly, we find that most effects of the editorial review sentiments (i.e., heterogeneous treatment effects of sentiment in the main specification) are insignificant or marginally significant, except those of *Food Course*. These results suggest that either editorial reviews are highly homogeneous in sentiments or the heterogeneity does not impose an additional impact on most dimensions of subsequent user reviews. The former explanation is consistent with the summary statistics shown in Table 4. The variation of the sentiment of editorial reviews is roughly one-third of the variation of the sentiment of user reviews. We further leverage an alternative identification scheme that controls for heterogeneity through matching. Specifically, we construct a compound sentiment score, following Hutto and Gilbert (2014), for editorial reviews and then keep only those restaurants with similar compound sentiment scores (i.e., in the range of average compound score ± 0.05) as an additional matching stage to erase the sentiment heterogeneity. We then use those matched restaurants for the main specification and mechanism analysis. As shown in Tables 17–20 in Online Appendix B, the effects of *EditorialReview* remain similar to the main results.

6.6. Alternative Matching Methods

In this subsection, we demonstrate that our results are not driven solely by the specific matching algorithm (i.e., the nearest neighbor matching) used in PSM. Here, instead, we adopt the entropy-balancing approach, which is widely used in the literature (e.g., Diestre et al. 2020, Jiang et al. 2020), and update the estimation results using the main DID regression specification. The results, shown in Tables 21 and 22 in Online Appendix B, are consistent with our main estimation results, indicating their robustness to matching algorithms. Also note that the results for mechanism analysis (shown in Tables 23 and 24 in Online Appendix B) are not sensitive to the matching approach.

7. Conclusions

Third-party reviews, such as those written by expert reviewers, have a strong impact on a consumer's purchasing decisions and a company's sales. Such an economic implication is also observed for consumer reviews. Although prior studies have shown how consumer reviews affect subsequent reviews, there is virtually no previous work that examines the influence of third-party reviews on consumer reviews. Our study addresses this gap in the literature by collaborating with a large restaurant review platform in Asia to investigate the impact of editorial reviews, a form

of third-party reviews, on subsequent consumer reviews both quantitatively and qualitatively.

By leveraging a quasiexperiment setup, we find empirical evidence that editorial reviews significantly influence subsequent user reviews and users' review-writing behavior. Specifically, editorial reviews increase the quantity of subsequent user reviews and reshape their textual content. Restaurants that have editorial reviews receive more consumer reviews that not only are longer but also, resemble the editorial review(s) in regard to the topic, sentiment/rating, length, and readability in the subsequent period. The increased length comes from augmented content on the topics that appear more in editorial reviews, whereas the content for the other topics does not decrease, suggesting a net increase in the content variety. The reviewer-level analysis for long-term reviewers further attributes the changes in their reviews to their behavioral changes under identifying assumptions. This finding suggests users' herding behavior toward platform editors in how to write reviews.

Our work makes a substantial contribution to academic literature and practice. From the perspective of academic research, we contribute to the extant literature on herding effects in the specific context of UGC by conceptualizing and identifying users' herding behaviors toward platform editors. Such a connection extends the scope of our understanding of peer influence on the UGC platform, as prior literature focuses only on the influence of peer users. It is particularly difficult, however, to extrapolate these prior findings and apply them to the context of editorial reviews because editorial reviews are significantly different from consumer reviews, and the platform control over these two types of feedback is also vastly different. In addition, we contribute to the literature on third-party reviews by revealing their additional economic value through examining their impact on WOM. This helps marketers to further comprehend the value of third-party reviews. Such an effect of third-party reviews is increasingly important because of the ubiquity of digitized UGC and eWOM in online platforms. In addition, our study contributes to the literature on the management of UGC by platforms, as our findings suggest the effectiveness of using an editorial review program to manage UGC in both quantitative aspects and qualitative aspects. We are among the first to demonstrate empirically the influence of editorial reviews on subsequent consumer reviews and reveal the underlying mechanisms of such an influence.

From a practical perspective, our research provides insights that could help platform managers enhance their reviewers' contributions and review content, both of which are crucial to the platform's competitive advantage and long-term sustainability. Herding by peer reviewers toward editors indicates that platforms could use editorial reviews to influence and manage

the content generation of its users. Because the platform can more exogenously manage the content of editorial reviews and determine for which businesses to provide editorial reviews as compared with consumer reviews, platform managers could leverage editorial reviews not only to encourage more users to write reviews for targeted businesses but also, to influence the topic, sentiment/rating, length, and readability of their written expressions. Specifically, high-quality editorial reviews could lead platform users to provide more content, cover desired topics more, and use a more neutral tone and readable writing style when contributing reviews to the platform. Such richer content would further reduce information seekers' cost of searching and improve their overall experience on the platform. Therefore, the managerial implications of our research are especially valuable for relatively mature review platforms that have accumulated enough review volume and have shifted their focus to the qualitative aspect of the content and the reading experience of users. Nevertheless, it is also important to note that an improper use of editorial reviews may adversely affect the platform. For example, businesses that receive editorial reviews tend to attain more reviews afterward. As such, if the platform only provides editorial reviews for popular businesses, it is possible that reviews on the platform may be concentrated on a small set of popular restaurants. Such a decrease in review diversity may negatively impact the platform's long-term sustainability.

Being one of the first works to investigate the interplay between editorial reviews and consumer reviews, our work is subject to several limitations, which also represent avenues for future research. First, the editorial/consumer reviews in our data set are exclusively for experience goods. It would be interesting to determine whether the effects we observe remain the same for reviews of search goods. Second, identifying herding effects is inherently challenging in observational data (Manski 1993) because we can isolate only such effects among long-term users under identifying assumptions. Further works that utilize different research methodologies, such as randomized experiments, may challenge such assumptions and reexamine herding effects among all users. Alternatively, follow-up surveys or small-group interviews may further reveal the users' states of mind and uncover why such changes occur. Third, the review platform in our study is located in Asia, and more than 95% of the users are from Asia. A future study could investigate whether herding/differentiating behavior differs in another culture. Finally, future research that has access to panel data of traffic and sales may be able to further attribute the increased volume of user reviews (i.e., herding in which restaurants to visit and subsequently, to review) to the awareness and conversion changes of users and to confirm whether the

indirect impact of editorial reviews on consumer reviews generates cascading effects on product sales.

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Endnotes

¹ Among 5,158 restaurants with advertising contracts, 958 have received editorial reviews (about 1/5). In addition, among 139,983 restaurants without advertising contracts, 1,325 have received editorial reviews (about 1/100).

² In Figure 1 in Online Appendix A, we also show that the user review differences before and after signing advertising contracts, visualized with the same approach, are not as salient, indicating that the advertising contracts may not confound dramatically the impact of editorial reviews on user reviews.

³ Few treatment restaurants are dropped in the PSM stage. Therefore, there is a minimal distribution addition for the treatment group in Figure 3.

⁴ For the categorical covariates included, we are unable to aggregate, present, and compare their values in a concise manner because the number of categories is particularly large. Note, however, that we apply exact matching so that the systematic difference between treatment and control groups is zero.

⁵ Throughout this paper, because the characteristic of each dependent variable is different, each model is estimated separately.

⁶ It is possible that editorial reviews for a restaurant are provided in the middle of time T . According to our current specification, the full month T is taken as the treatment, which might result in an underestimated effect of the treatment. As such, we show the robustness of its qualitative interpretation in Section 6.2.

⁷ Note that the analysis for review volume is at an aggregate (i.e., not individual) level and that the reviews for the treatment and control restaurants at any point in time come from the same set of platform users who have the same aggregate users' homophily (i.e., inherent taste). Hence, the effect of *EditorialReview* on volume is not specifically driven by homophily.

⁸ $S_{ikt}^{UR,neg}$ must be dropped to avoid collinearity.

⁹ Note that adding these two variables results in dropping each restaurant's first user review, which does not have $PriorVol_{ikt}$ and $PriorRate_{ikt}$. To show robustness, we also estimate with the full observation but without controlling for $PriorVol_{ikt}$ and $PriorRate_{ikt}$, as seen in Online Appendix C.1. The estimated coefficients for *EditorialReview* remain similar.

¹⁰ In Online Appendix C.2, we show that an alternative specification without a bootstrap process, where an additional control variable *ReviewVolume* is added, leads to qualitatively similar results.

¹¹ These data are also bootstrapped as we work on the review-level analysis.

¹² We empirically demonstrate the effect of editorial reviews on the approximate number of customers of treated restaurants in Online Appendix C.5.

¹³ As an alternative specification, we also consider the specification where $\Delta Length_{ikt}$ and $\Delta Rating_{ikt}$ are calculated without an absolute value. We report this result in Online Appendix C.4.

¹⁴ As a consequence, each restaurant's first review, which has no prior reviews, must be dropped, and we see a decreased number of observations.

¹⁵ We also apply this alternative specification to the mechanism analysis in Online Appendix C.1. The estimation results, shown in Tables 27 and 28 in Online Appendix C, remain qualitatively the same.

¹⁶ *PositiveScore* is dropped to avoid collinearity.

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